

# Blockchain-Based Certificate-Free Cross-Domain Authentication Mechanism for Industrial Internet

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**Abstract**—In Industrial Internet, mutual authentication between enterprises is a prerequisite for establishing reliable upstream and downstream relationships. Existing authentication methods suffer from complicated certificate management and key escrow problems. Moreover, many authentication mechanisms cannot resist common security attacks and have high computational overhead and communication costs. Therefore, this article proposes a blockchain-based certificate-free cross-domain authentication mechanism for Industrial Internet. By establishing an Ethereum consortium blockchain as the trusted cornerstone among different regions, industrial enterprises in each region generate the user's private key with the key generation center in the region, thus avoiding the key escrow problem. This consortium blockchain adopts the Proof-of-Authority consensus mechanism for scalability and throughput. Industrial enterprises in different regions invoke smart contracts and query other industrial enterprises for mutual authentication and key negotiation. SVO logic proves the proposed scheme achieves the intended authentication goal, and the automated formal verification tool Scyther proves the scheme's security. In addition, compared with seven related schemes in the last three years, the experimental results show that the proposed scheme has low communication overhead and computational cost in the authentication key negotiation phase. The experiments on the Ethereum consortium blockchain built by Raspberry Pi prove the effectiveness of the proposed scheme. Finally, the comparative analysis of common security properties proves the reliability of the scheme.

**Index Terms**—Authentication, blockchain, certificate-free, Industrial Internet.

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## I. INTRODUCTION

INDUSTRIAL Internet is an essential pillar in promoting the rapid development of the industry. With increasingly intelligent terminal devices linked to the Industrial Internet, the Industrial Internet needs to process hundreds of millions of data daily. Due to the confidentiality requirements of all industrial enterprises for their data, most companies are willing to their data private. Boundaries exist between the systems of different enterprises, thus causing the problem of data silos [1]. In addition, the data obtained by industrial enterprises from various sensor devices, automation devices, and intelligent production devices are multisource heterogeneous [2]. How to mine these data to improve data utilization is also an urgent problem for enterprises. Generally, there are distances between the upstream and downstream of enterprises. Most large industrial enterprises distribute themselves across network regions. Many industrial enterprises exist in the same network region. However, the cost of data interoperability among industrial enterprises in different domains is high. The lack of trust among the participants makes direct interaction a significant security risk.

Cross-domain authentication and key negotiation are essential to solving the above problem [3]. For industrial enterprises in different domains, if two parties need to share data, they must first prove to each other that they are legitimate and trustworthy enterprises. When two parties need to conduct a large amount of confidential data transmission, they need to negotiate the key for data encryption in advance, i.e., key negotiation. After conducting cross-domain negotiation of keys, the two parties transmit data securely with the negotiated keys. Most existing authentication methods require a certificate [4]. Each enterprise organization applies for a certificate from a certificate authority center (CA), and then performs cross-domain authentication and key negotiation based on the certificate's signature from the CA. However, the method suffers from complex certificate management and single-point-of-failure problems.

In addition, the identity-based cross-domain authentication method has the key escrow problem [5], i.e., a private key generator generates a private key for a given identity and has the power to decrypt all ciphertexts. Suppose hackers compromise the private key generator. In that case, they can not only decrypt the current user's information but also view the user's previous information, as the private key generation center (KGC) knows the private keys of all users. So the method

cannot yet achieve forward confidentiality. By absorbing the fast computation speed of identity-based cryptosystem, Al-Riyami and Paterson [6] proposed a certificate-free cryptographic mechanism in which the user and the KGC jointly generate the user's private key. Therefore, this mechanism can solve the key escrow problem and the certificate management complexity. However, these cryptographic mechanisms must know each other's public keys before authentication. Suppose there is no reliable public platform to store the public keys. The entities will be vulnerable to identity impersonation attacks.

This article aims to establish reliable trust management among different industrial enterprises. It can achieve cross-domain authentication of different industrial enterprises. Additionally, it can make the certificate-free cryptographic mechanism effectively resist the public key replacement attack and reduce the computation. As blockchain technology gradually matures [7], it becomes a good authentication platform [8] due to its characteristics, such as immutability [9], [10] and multiparty consensus [11], [12]. Therefore, this article proposes a blockchain-based certificate-free cross-domain authentication mechanism for Industrial Internet. First, This article constructs an Ethereum consortium blockchain among industrial enterprises distributed in different regions. Then, a KGC is set up in each domain, responsible for generating partial private keys of industrial enterprises in the region. Each industrial enterprise generates its complete private key according to the received partial private key. The KGC and the industrial enterprise upload their public keys on the consortium blockchain. When an industrial enterprise wants to negotiate business with industrial enterprises in other regions across domains, it needs to prove its legitimacy to the target industrial enterprise. Finally, the industrial enterprises in the two regions negotiate their data-sharing keys.

#### A. Motivation

Most existing authentication methods adopted certificate-based authentication, which suffers from the complexity of certificate management and the inability to find a mutually recognized certification authority for industrial enterprises far from each other. Identity-based authentication has key escrow problems, and the computational and communication costs of identity-based authentication mechanisms are high. Blockchain-based authentication mechanisms have more security issues. In conclusion, the existing authentication mechanisms in the Industrial Internet environment have high computational and communication costs in the authentication key negotiation process. They can resist fewer attacks and need more advanced formal analysis tools to verify them. Therefore, this article proposes to design a blockchain-based cross-domain authentication mechanism adapted to the Industrial Internet and prove its security by SVO logical proof and formal analysis tool Scyther.

#### B. Contribution

In order to address the fundamental limitations currently in the related work, the design decision of the proposed scheme is to achieve lightweight cross-domain authentication

by combining a certificate-free cryptosystem with a consortium blockchain. The consortium blockchain mainly stores industrial companies' identity and public key to ensure their immutability and publicness. The signature generation in the proposed scheme must incorporate information, such as the sender's private key, the receiver's public key, and random numbers to resist forgery attacks. The verifier must know the private key of the receiver and the public key of the sender in order to decrypt the message and verify the signature. The sender must receive a response after sending the message, and the response must contain the sender's temporary new random value to resist the man-in-the-middle and replay attacks. The SVO logical proof verifies that the proposed scheme achieves the authentication goal. The proposed scheme continuously tests the scheme's security with the help of automated formal analysis tools until finding no attacks. Thus, the main contributions of this article include the following.

- 1) This article proposes a cross-domain mutual authentication mechanism for the Industrial Internet scenario based on the Decisional Diffie-Hellman (DDH) difficulty problem and symmetric encryption algorithm, which achieves the intended authentication goal under the SVO logical proof. The proposed mechanism can not only solve the key escrow problem but also avoid complex certificate management, thus reducing the storage overhead of the authentication system.
- 2) This article innovatively builds an Ethereum consortium blockchain based on the Proof-of-Authority (PoA) consensus mechanism in the Raspberry Pi environment. The experimental results prove the proposed scheme can apply in the Industrial Internet environment. The consortium blockchain has not only good scalability but also high throughput.
- 3) This article constructs a key negotiation mechanism on a consortium blockchain. The automated security protocol verification tool Scyther proves the security of the proposed scheme, which can resist multiple types of attacks while satisfying specific security properties.
- 4) Compared with seven related schemes from the last three years, simulation experiment results demonstrate the proposed scheme's low computational overhead and communication cost in the authentication key negotiation phases. The proposed scheme has more high security in terms of security properties.

#### C. Road Map

The organization of the remainder of this article is as follows. Section II presents the related research work. Section III lists the preparatory work on the authentication mechanism. Most importantly, in Section IV, details of the implementation of the proposed scheme are presented. To prove the security of the proposed scheme, Section V presents a formal proof and automated formal analysis of the scheme and describes the security properties that the scheme possesses. Section VI performs an experimental comparative analysis. Finally, Section VII summarizes this article and gives future work.

TABLE I  
SUMMARY OF RELEVANT AUTHENTICATION MECHANISMS

| Mechanisms                 | Schemes          | Advantages                                                                 | Disadvantages                                                                  |
|----------------------------|------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Certificate based          | [13], [14], [15] | Computationally efficient                                                  | Complex certificate management                                                 |
| ECC based                  | [16], [17], [18] | Computationally efficient                                                  | Key escrow problem                                                             |
| Identity based             | [19], [20]       | Cross-domain authentication                                                | High computational and communication costs                                     |
| Certificate-free based     | [21]             | Support for public key revocability;<br>No need for certificate management | High computational and communication costs;<br>Unable to resist forgery attack |
|                            | [22]             | Computationally efficient;<br>No need for certificate management           | Unable to resist man-in-the-middle attack                                      |
| Blockchain based           | [23]             | Cross-domain authentication                                                | Unable to resist man-in-the-middle attack                                      |
|                            | [24]             | Decentralized Authentication                                               | Unable to resist forgery attack                                                |
|                            | [25]             | Decentralized Authentication                                               | High computational and communication costs                                     |
|                            | [26]             | Cross-domain authentication                                                | High communication overhead                                                    |
|                            | [27]             | Decentralized Authentication                                               | Key compromise risk                                                            |
| Zero knowledge proof based | [28]             | Cross-domain authentication                                                | Reliance on certificate authority                                              |

## II. RELATED WORK

Presently, authentication protocols mainly include five categories: 1) certificate-based authentication protocols; 2) elliptic curve cryptography (ECC)-based authentication protocols; 3) identity-based authentication protocols; 4) certificate-free authentication protocols; and 5) blockchain-based authentication protocols. Table I summarizes the relevant authentication protocols, including the types, advantages, and disadvantages of the relevant authentication schemes. Each type of related scheme is described in detail below.

Certificate-based authentication protocols rely on public-key infrastructure, where each node joining the network must obtain a digital certificate issued by a CA. The nodes in the network authenticate each other by examining the digital certificate. For example, to solve the problems such as delay in the Diffie–Hellman protocol in wireless sensor networks, Sun et al. [13] proposed a fully authenticated certificate-based Diffie–Hellman protocol. Lin et al. [14] designed a conditional privacy-preserving authentication scheme based on the elliptic curve digital signature algorithm in public-key infrastructure to solve the problem of storing many keys for in-vehicle units and related organizations in vehicular networking. The scheme aims to achieve secure and efficient authentication in vehicular communication while reducing the key storage overhead. Garba et al. [15] designed an Ethereum-based domain authentication scheme to prevent the problem of CAs abusing their authority to issue certificates for domain names. In the scheme, clients query a set of trusted CAs recorded in the blockchain to request trusted CAs to issue certificates for themselves. With the increase in the number of nodes, the certificate-based authentication method needs to manage many certificates and thus suffers from the problem of certificate management complexity.

Smart grids widely use ECC-based authentication protocols. For example, Garg et al. [16] designed a secure and lightweight authentication scheme that uses a fully hashed menezes-qu-vanstone key exchange mechanism and ECC adapted to smart grids. The authentication scheme is suitable for establishing secure communication between smart meters (SMs) and neighborhood area network (NAN) gateways in an insecure channel.

However, Chaudhry et al. [17] argued that Garg et al.'s scheme is not resistant to key leakage impersonation attacks and does not have forward secrecy. So they proposed a secure and efficient authentication scheme to achieve device-to-device authentication. Chaudhry et al. [18] proposed another lightweight authentication scheme based on ECC to address the impersonation, SM traceability, and SM physical capture attacks between SM and NAN. However, the scheme still does not solve the key escrow problem that exists in the trust center.

Identity-based authentication protocols are designed based on identity-based cryptosystems, which do not require managing many certificates because the user's identity can act as the user's public key in identity-based cryptosystems. The method is computationally efficient and well suited for client–server mode. For example, Shen et al. proposed a blockchain-assisted secure device authentication mechanism (BASA) to deal with the heavy key management overhead of the existing authentication methods. BASA is a cross-domain industrial IoT authentication mechanism based on identity signature and verification algorithms with the help of a blockchain platform [19]. In order to achieve efficient authentication of sensors by cloud servers, Xiong et al. [20] constructed an identity-based authentication protocol between cloud servers and sensors. However, since the private key generator knows the private keys of all users, the identity-based authentication method suffers from the key escrow problem and the single point of failure problem.

Certificate-free cryptosystem applied to authentication protocols can reduce the storage overhead of nodes that can prevent malicious behavior of KGCs [6]. For example, Xiong and Qin [21] proposed a certificate-free anonymous remote authentication protocol based on a binomial tree that supports reversibility. It aims to prevent expired users from continuing to enjoy services provided by application service providers for free and malicious application service providers from accessing users' health privacy data for commercial interests. However, Shim [29] demonstrated that the certificate-free signature scheme in that paper can not resist the attack that an adversary who knows the secret value forges the signature of any message in the same period. Therefore, this

authentication protocol cannot achieve the purpose of mutual authentication. Deng and Gao [22] proposed a key negotiation scheme with certificate-free bidirectional authentication for smart grids to communicate securely between SMs and utilities. However, the scheme is not resistant to man-in-the-middle attacks.

Blockchain-based authentication methods leverage the blockchain's decentralization [30], immutability [31], and programmability to achieve decentralized authentication. For example, Feng et al. [23] proposed a blockchain-based cross-domain authentication scheme for drones to address the communication problem between drones in different domains. They achieved this by deploying a private blockchain to manage drone device registration, authentication, and auditing in the local domain. To achieve efficient authentication between different IoT platforms, Guo et al. [24] built a distributed authentication system relying on smart contracts for name resolution and ECC to ensure communication security with an edge computing platform. In order to solve the single point of failure problem of traditional centralized authentication for IoT, Cui et al. [25] established a hybrid distributed authentication system by building a private blockchain in a single wireless sensor network and linking all base station nodes of the wireless sensor network to the public blockchain. However, the scheme builds an authentication model based on an elliptic curve digital signature algorithm and lacks a formal security analysis process. Currently, the scheme is only conceptual and lacks experimental simulations.

To reduce unnecessary reauthentication during frequent handovers between heterogeneous cells in 5G networks, Yazdinejad et al. [26] designed a new secure authentication handover scheme using blockchain and software-defined networking, which is efficient in terms of privacy protection and attack resistance. However, Wang et al. [32] argued that the communication overhead between the software-defined networking controller and the blockchain center of this scheme is high. Yazdinejad et al. [27] proposed a decentralized blockchain-based authentication scheme for distributed patients in a hospital network. However, Liu et al. argue that the disadvantage of this scheme is that it is insecure for hospitals to share patients' public and private keys to decrypt their medical data. So the scheme has the risk of key leakage [33].

In addition, there are authentication mechanisms based on the combination of zero-knowledge proofs and smart contracts. For example, to improve the responsiveness of blockchain, Chen et al. [28] proposed a solution that uses the interplanetary file system as the storage layer for cross-domain data. They use blockchain as the control layer to ensure data consistency and integrity and leverage zero-knowledge proof algorithms in the application layer to achieve efficient cross-domain authentication.

The main issues include the following through the analysis and summary of the previous work. The certificate-based authentication mechanism has complicated certificate management. The ECC-based authentication scheme mainly suffers from key escrow problems. The identity-based authentication mechanism has high computational and communication costs. The certificate-free authentication mechanism is not resistant

TABLE II  
SYMBOL LIST

| Notation     | Description                                        |
|--------------|----------------------------------------------------|
| $G_1$        | Cyclic addition group                              |
| $G_T$        | Cyclic multiplication group                        |
| $\hat{e}$    | Bilinear mapping                                   |
| $Z_p^*$      | $\{x : 0 < x < p \text{ and } \gcd(x, p) = 1\}$    |
| $\{0, 1\}^*$ | A binary string of arbitrary length                |
| $H$          | Hash function                                      |
| $\sigma_X$   | Partial private key of X                           |
| $SX$         | Complete private key of X                          |
| $PX$         | Public key of X                                    |
| $? =$        | Judge whether values are equal                     |
| $\{X\}_K$    | AES symmetric encryption of X with key K           |
| $[X]_K$      | Digital signature message X with key K             |
| $*n$         | Unrecognized received messages                     |
| $Axn$        | Axioms in SVO logic, n represents the axiom number |
| $MP$         | Modus Ponens inference rule in SVO logic           |
| KDF          | Key derivation function                            |

to forgery attacks and man-in-the-middle attacks. Blockchain-based authentication mechanisms have more security issues. The authentication mechanism based on zero-knowledge proof is overly dependent on CAs. By absorbing the advantages of blockchain-based and certificate-free cryptosystems and avoiding the use of bilinear pairs as much as possible to reduce the computational cost and communication overhead, the proposed scheme can achieve cross-domain authentication and solve the key escrow problem and certificate management complexity.

### III. PRELIMINARY

This section describes the relevant mathematical notation definitions and difficult problems. It defines the functionality and trust levels of the system participants. Then, this article gives a specific application scenario for the scheme. Table II presents the relevant representational meanings of the notations used in this article.

#### A. Hard Problem

**Theorem 1 (DDH Problem):** Given a cyclic additive group  $G_1$  of order  $p$  and its generator is  $g$ . The DDH problem [34] in  $G_1$  is to determine whether  $c \equiv a \cdot b \pmod{p}$  holds given  $(g, a \cdot g, b \cdot g, c \cdot g)$ , where  $a, b, c \in Z_p^*$  are unknown integers.

#### B. System Model

This section analyzes the functions and trust levels of the participants in the Industrial Internet. It mainly consists of three entities: 1) KGC; 2) industrial enterprises; and 3) consortium blockchain.

- 1) *KGC*: It is a semi-trusted entity whose primary responsibility is to generate partial private keys of the regional industrial companies and participate in the consortium blockchain's consensus process.
- 2) *Industrial Enterprise*: It generates the complete private key and public key by leveraging the partial private keys generated by the local KGC for itself and publishes its



public key and identity information to the blockchain. Industrial enterprises in other organizations retrieve that enterprise's information and public key in the blockchain to authenticate and negotiate the key with it and perform subsequent data transfer.

- 3) *Consortium Blockchain*: The Ethereum consortium blockchain has various organizations working together to build it. Among them, KGC is responsible for the PoA consensus [35] process of the consortium blockchain and backing up the blockchain data.

### C. Authentication Goals

This article will apply the more mature SVO logic [36] to prove whether the authentication key negotiation mechanism achieves the intended authentication goals. Reference [37] includes the meaning of the mathematical symbols involved, and the authentication goals of the proposed scheme mainly include the following.

- 1) *Alive Confirmation*:  $A$  believes  $B$  says  $X$ .
- 2) *Entity Authentication*:  $A$  believes  $B$  says  $F(X, Na)$ .
- 3) *Secure Key Establishment*:  $A$  believes  $A \xleftrightarrow{K^-} B$ .
- 4) *Key Confirmation*:  $A$  believes  $A \xleftrightarrow{K^+} B$ .
- 5) *Key Freshness*:  $A$  believes  $\text{fresh}(K)$ .
- 6) *Mutual Understanding of Shared Key*:  
 $A$  believes  $B$  says  $B \xleftrightarrow{K^-} A$ .

### D. Application Scenarios

High-energy-consuming industries can apply the proposed scheme to the low-carbon trading market [38], [39]. Fig. 1 shows a blockchain-based authentication model for low-carbon transactions in the steel industry. Iron ore is first mined from the mine and transported to the beneficiation enterprise. The beneficiation enterprise sells the ore to the steel enterprise after rough processing. The steel enterprise uploads the carbon emission data related to the manufacturing process to the blockchain after a series of operations, such as refining and purification. The downstream manufacturing industries, such as shipbuilding and automobile manufacturing plants, select low-carbon steel enterprise products according to the carbon emission data of steel enterprises. Then, steel and manufacturing enterprises in different regions must carry out mutual authentication [40] and key negotiation first. Then, the two parties negotiate business data, and the contract agreed upon by the two parties can also use smart contracts to upload hash values to the blockchain storage to prevent repudiation and build a low-carbon society.

## IV. CERTIFICATE-FREE CROSS-DOMAIN AUTHENTICATION MECHANISM

This section details the certificate-free authentication mechanism of this article, which consists of three main parts: 1) the system initialization phase; 2) the industrial enterprise registration phase; and 3) the industrial enterprise authentication and key negotiation phase.

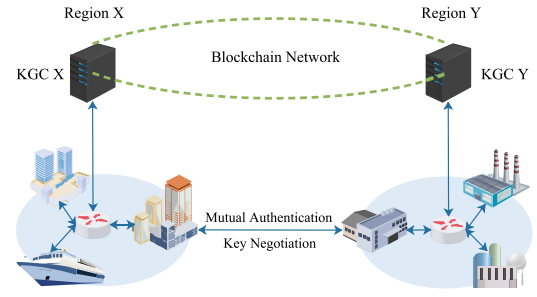


Fig. 1. Blockchain-based authentication model for low-carbon transactions in the steel industry.

### A. Initialization Phase

During this phase, industrial enterprises construct an Ethereum consortium blockchain among regions by dividing them based on their geographical location or network structure. Each region is an organization, and each organization contains many industrial enterprises. There is one KGC per organization, and the KGC is responsible for registering different industrial enterprises. The consortium blockchain selects a KGC responsible for the initialization operation of the system. The KGC chooses an additive group  $G_1$ , defines the generator and order of the group  $G_1$  as  $g$  and  $p$ , respectively, a hash function  $H_1 : \{0, 1\}^* \rightarrow Z_p^*$  and another hash function  $H_2 : \{0, 1\}^* \rightarrow \{0, 1\}^{256}$ . The KGC uploads the system parameters  $params = \{G_1, p, g, H_1, H_2\}$  to the blockchain. In the initial stage of the creation of the Ethereum blockchain, the KGC that generates the system parameters writes a smart contract and uploads it to the blockchain. This smart contract mainly consists of two functions, one function stores information, such as the identity and public key of the user or entity, and the other function queries the identity and public key of the user and entity. Therefore, industrial companies can query the identity and public key of other users and other information by calling the smart contract on the blockchain. The KGC in each organization sets its public key by querying system parameters in the blockchain system and uploads its public key and identity information to the blockchain system.

### B. Registration Phase

The registration stages of industrial enterprises in each organization are identical. This article takes the example of industrial enterprise A applying for registration to the KGC in organization X, as shown in Fig. 2. First, the KGC of organization X randomly selects an element from  $Z_p^*$  as its private key  $PRX$ , i.e.,  $PRX \in Z_p^*$ . Then, it generates the public key  $PKX = PRX \cdot g$  for the KGC of organization X and uploads its public key to the blockchain. When industrial enterprise A in organization X registers, it sends its identity information to the KGC of organization X. This KGC randomly selects an element  $Ar_2$  from  $Z_p^*$  and calculates  $AR_2 = Ar_2 \cdot g$  and  $Ah_1 = H_1(IDA)$ . It generates a signature  $\sigma_A$  on industrial enterprise A and sends  $\sigma_A$  and  $AR_2$  to industrial enterprise A, where  $\sigma_A = PRX + Ar_2 \cdot Ah_1$ . Industrial enterprise A selects an element  $Ar_1$  from  $Z_p^*$  and calculates  $AR_1 = Ar_1 \cdot g$ . Industrial enterprise A generates its private key  $SA = \sigma_A + Ar_1$  and

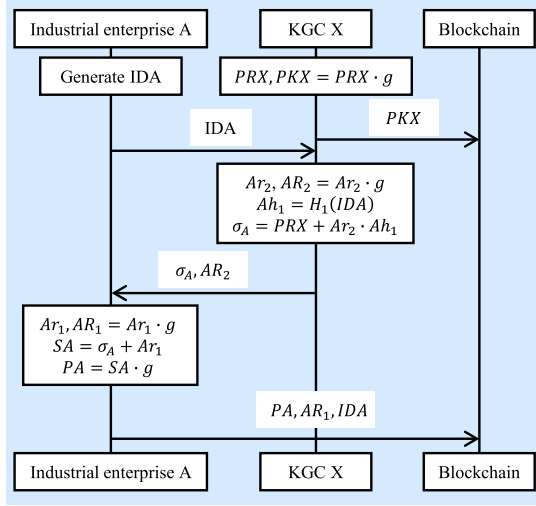


Fig. 2. Industrial enterprise A applies for registration to organization X sequence diagram.

public key  $PA = SA \cdot g$ . Finally, industrial enterprise A sends its public key  $PA$ , identity  $IDA$ , and  $AR_1$  onto the blockchain to facilitate the retrieval and contact of industrial enterprises from other organizations.

When industrial enterprise A receives the partial private key sent to itself by the KGC from its organization, it will verify whether the received partial private key is correct by (1). As each organizational KGC participates in the consensus process of the blockchain, the KGC in this organization broadcasts  $AR_2$ , which enables other participating nodes to verify that industrial enterprise's public key. Thus, the correctness verification process of the public key of the industrial enterprise is shown in (2). In this formula, other verification nodes can verify which organization the enterprise belongs to and can verify the legitimacy and correctness of the public key based on the  $AR_2$  provided by the KGC of the organization

$$\sigma_A = PRX + AR_2 \cdot Ah_1 \quad (1a)$$

$$\sigma_A \cdot g = PRX \cdot g + AR_2 \cdot Ah_1 \cdot g \quad (1b)$$

$$\sigma_A \cdot g = PKX + AR_2 \cdot Ah_1 \quad (1c)$$

$$\begin{aligned} PA &= SA \cdot g \\ &= (\sigma_A + AR_1) \cdot g \\ &= (PRX + AR_2 \cdot Ah_1 + AR_1) \cdot g \\ &= PKX + Ah_1 \cdot AR_2 + AR_1. \end{aligned} \quad (2)$$

### C. Authentication Key Negotiation Phase

Fig. 3 shows the authentication key negotiation phases of industrial enterprise A in organization X and industrial enterprise B in organization Y. First, industrial enterprise A looks up the public key  $PB$  and identity  $IDB$  of industrial enterprise B from the consortium blockchain to communicate. Similarly, industrial enterprise B looks up the public key  $PA$  and identity  $IDA$  of industrial enterprise A from the consortium blockchain. Industrial enterprise A randomly generates a random number  $Ar_3$  and timestamp  $t_1$ , where  $Ar_3 \in \mathbb{Z}_p^*$ , and

computes  $rpb = Ar_3 \cdot PA$ ,  $Ah_3 = H_1(IDA, IDB, Ar_3, t_1)$ , and  $QAB = SA \cdot (Ar_3 + Ah_3)$ . Then, industrial enterprise A sends  $rpb$ ,  $Ah_3$ ,  $QAB$ , and  $t_1$  to industrial enterprise B. Industrial enterprise B randomly generates a random number  $Br_3$  after receiving the communication request, where  $Br_3 \in \mathbb{Z}_p^*$ . Then, industrial enterprise B can determine whether the information sent by industrial enterprise A is correct according to (3), its own private key  $SB$  and public key  $PA$  of industrial enterprise A. Since only industrial enterprise B knows its private key, it completes the authentication of industrial enterprise A and generates the key  $sk_{ab} = Br_3 \cdot SB \cdot rpb$  for key negotiation between the two parties at the same time. Then, industrial enterprise B generates a new timestamp  $t_2$  and calculates  $Bh_3 = H_1(IDA, IDB, Br_3, t_1, t_2)$ ,  $QBA = SB \cdot (Br_3 + Bh_3)$ ,  $rpa = Br_3 \cdot PB$ . Subsequently, industrial enterprise B sends  $rpa$ ,  $Bh_3$ ,  $QBA$ ,  $t_2$  to Industrial enterprise A. Industrial enterprise A can determine whether the information sent by industrial enterprise B is correct according to (4),  $Ar_3$ , and  $PB$ . Since only industrial enterprise A knows its private key, thus completing the authentication of industrial enterprise B. Finally, two industrial enterprises determine whether the other party has generated the same negotiation key by encrypting the message  $H_2(QAB, QBA, Ah_3, Bh_3, t_1, t_2)$  with their own generated negotiation key

$$\begin{aligned} &QAB \cdot Br_3 \cdot PB \\ &= SA \cdot (Ar_3 + Ah_3) \cdot Br_3 \cdot PB \\ &= SA \cdot Ar_3 \cdot Br_3 \cdot PB + SA \cdot Ah_3 \cdot Br_3 \cdot PB \\ &= Ar_3 \cdot Br_3 \cdot SA \cdot PB + Ah_3 \cdot Br_3 \cdot SA \cdot PB \\ &= Ar_3 \cdot Br_3 \cdot PA \cdot SB + Ah_3 \cdot Br_3 \cdot PA \cdot SB \\ &= rpb \cdot Br_3 \cdot SB + Ah_3 \cdot Br_3 \cdot PA \cdot SB \end{aligned} \quad (3)$$

$$\begin{aligned} &QBA \cdot Ar_3 \cdot PA \\ &= SB \cdot (Br_3 + Bh_3) \cdot Ar_3 \cdot PA \\ &= SB \cdot Br_3 \cdot Ar_3 \cdot PA + SB \cdot Bh_3 \cdot Ar_3 \cdot PA \\ &= Br_3 \cdot Ar_3 \cdot SB \cdot PA + Bh_3 \cdot Ar_3 \cdot SB \cdot PA \\ &= Br_3 \cdot Ar_3 \cdot PB \cdot SA + Bh_3 \cdot Ar_3 \cdot PB \cdot SA \\ &= rpa \cdot Ar_3 \cdot SA + Bh_3 \cdot Ar_3 \cdot PB \cdot SA. \end{aligned} \quad (4)$$

## V. FORMAL VERIFICATION AND SECURITY ANALYSIS

This section will perform formal verification and security analysis of the proposed scheme. First, the proposed scheme achieves the intended authentication goal through SVO logic proof. Second, the automated formal analysis software Scyther [41] proves the security of the proposed scheme, including the formal analysis of the registration and authentication key negotiation phases. Finally, this section provides a detailed analysis of the security properties of the proposed scheme.

### A. Logical Analysis of Authentication Key Negotiation Mechanism

This section uses SVO logic to prove whether this article's authentication key negotiation mechanism achieves the intended authentication goal.

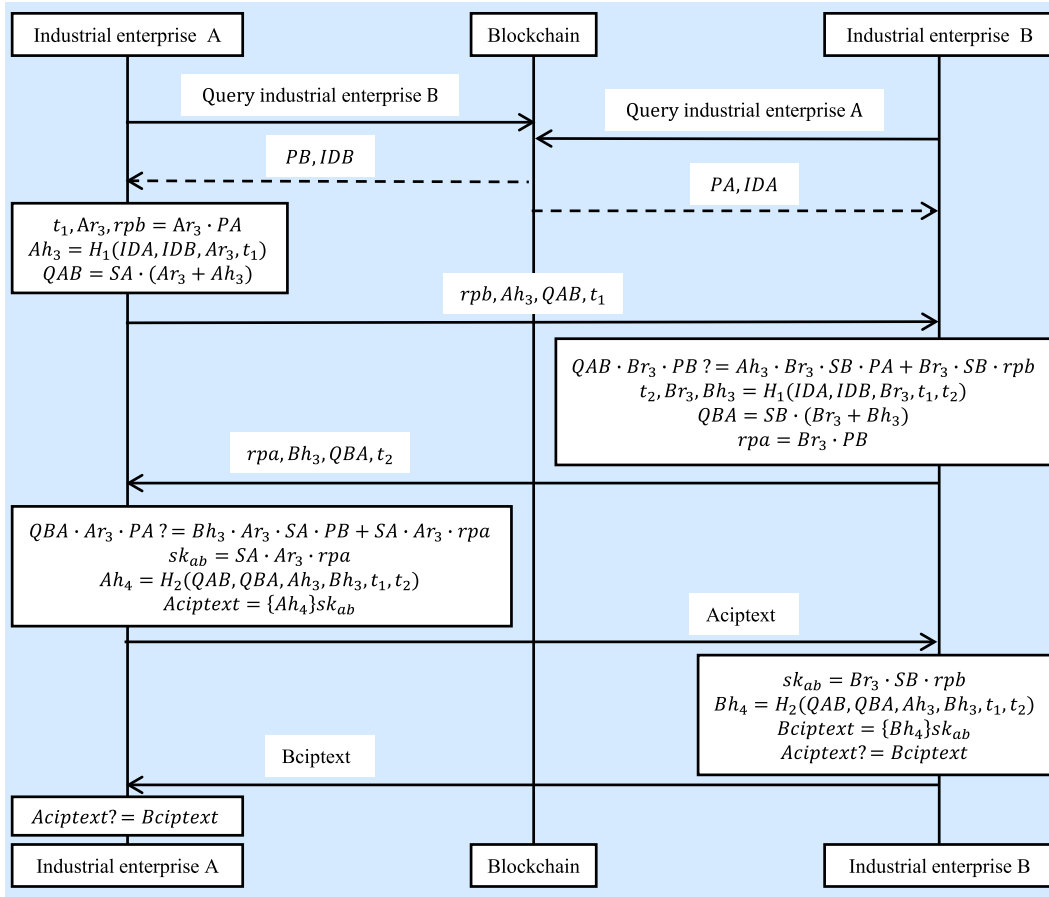


Fig. 3. Industrial enterprise mutual authentication key negotiation sequence diagram.

*Proof:* The SVO logic includes a total of four types of prerequisite assumptions. The first type of assumption is the initialization assumption

$$A \text{ believes fresh}(t_1), \quad B \text{ believes fresh}(t_2) \quad (5)$$

$$A \text{ believes fresh}(Ar_3), \quad B \text{ believes fresh}(Br_3) \quad (6)$$

$$A \text{ believes } PK(B, PB), \quad B \text{ believes } PK(A, PA). \quad (7)$$

The second type of assumption reflects the messages received during the protocol's operation

$$A \text{ received } (rpa, Bh_3, QBA, t_2) \quad (8)$$

$$B \text{ received } (rpb, Ah_3, QAB, t_1) \quad (9)$$

$$A \text{ received } Bciphertext$$

$$B \text{ received } Aciphertext. \quad (10)$$

The third type of assumption is the premise which reflects the degree of understanding of the information received by each subject

$$A \text{ received } (*_1, Bh_3, *_2, t_2) \quad (11)$$

$$B \text{ received } (*_3, Ah_3, *_4, t_1) \quad (12)$$

$$A \text{ received } *_5 \quad (13)$$

$$B \text{ received } *_6. \quad (14)$$

The fourth type of assumption reflects the premise of the receiver's interpretation of the information received

$$A \text{ believes } (A \text{ received } (*_1, Bh_3, *_2, t_2) \supset$$

$$A \text{ received } (\{Br_3\}_{PB}, H_1(IDA, IDB, Br_3, t_1, t_2), [Br_3, Bh_3]_{SB}, t_2)) \quad (15)$$

$$B \text{ believes } (B \text{ received } (*_3, Ah_3, *_4, t_1) \supset$$

$$B \text{ received } (\{Ar_3\}_{PA}, H_1(IDA, IDB, Ar_3, t_1), [Ar_3, Ah_3]_{SA}, t_1)) \quad (16)$$

$$A \text{ believes } (A \text{ received } *_5 \supset A \text{ received } (\{H_2(QAB, QBA, Ah_3, Bh_3, t_1, t_2)\}_{sk_{ab}},$$

$$A \xleftrightarrow{sk_{ab}} B, \text{ fresh}(sk_{ab}))) \quad (17)$$

$$B \text{ believes } (B \text{ received } *_6 \supset B \text{ received } (\{H_2(QAB, QBA, Ah_3, Bh_3, t_1, t_2)\}_{sk_{ab}},$$

$$A \xleftrightarrow{sk_{ab}} B, \text{ fresh}(sk_{ab}))). \quad (18)$$

Next, the logical derivation of the message received by industrial enterprise A begins

$$A \text{ believes } A \text{ received } (\{Br_3\}_{PB}, H_1(IDA, IDB, Br_3, t_1, t_2), [Br_3, Bh_3]_{SB}, t_2) \text{ by (10), (12), Ax1, MP} \quad (19)$$

$$A \text{ believes fresh}(Bh_3) \text{ By (16), (5), Ax7, Ax17, Ax1, MP} \quad (20)$$

$A$  believes  $B$  said  $(Br_3, Bh_3)$   
By (16), Ax7, Ax9, (7), Ax1,  $MP$  (18)

$A$  believes  $B$  says  $(Br_3, Bh_3)$   
By (17), (18), Ax17, Ax19, Ax1,  $MP$  (19)

$A$  believes  $A$  received  $(A \xleftrightarrow{sk_{ab}} B, \text{fresh}(sk_{ab}), \{H_2(QAB, QBA, Ah_3, Bh_3, t_1, t_2)\}_{sk_{ab}})$   
By (11), (14), Ax1,  $MP$  (20)

$A$  believes  $\text{fresh}(H_2(QAB, QBA, Ah_3, Bh_3, t_1, t_2))$   
By (17), Ax17, Ax18, Ax1,  $MP$  (21)

$A$  believes  $A \xleftrightarrow{sk_{ab}^-} B$   
By (16), (7), Ax5, Ax1,  $MP$  (22)

$A$  believes  $A \xleftrightarrow{sk_{ab}^+} B$   
By (20), (21), (22), Ax7, Ax1,  $MP$  (23)

$A$  believes  $\text{fresh}(sk_{ab})$   
By (23), (6), Ax17, Ax18, Ax1,  $MP$  (24)

$A$  believes  $B$  said  $H_2(QAB, QBA, Ah_3, Bh_3, t_1, t_2)$   
By (20), Ax7, (23), Ax3, Ax1,  $MP$  (25)

$A$  believes  $B$  says  $H_2(QAB, QBA, Ah_3, Bh_3, t_1, t_2)$   
By (21), (25), Ax19, Ax1,  $MP$  (26)

$A$  believes  $B$  sees  $B \xleftrightarrow{sk_{ab}^-} A$   
By (20), (23), Ax3, Ax1,  $MP$ . (27)

Therefore, (19), (26), (22), (23), and (24) prove the alive confirmation, entity authentication, security key establishment, key confirmation, and key freshness of industrial enterprise B, respectively, where  $Bh_3 = H_1(IDA, IDB, Br_3, t_1, t_2)$ . To prevent the negotiation key from being eavesdropped during the network transmission, the proposed scheme achieves the purpose that industrial enterprise A believes industrial enterprise B has seen the final negotiation key according to (27). Industrial enterprise B encrypts a message with the unconfirmed negotiation key and sends the ciphertext to industrial enterprise A instead of achieving mutual key confirmation purposes. Since the proposed scheme is a symmetric protocol, industrial enterprise B also achieves the alive confirmation, entity authentication, security key establishment, key confirmation, and key freshness of industrial enterprise A. Similarly, for the mutual key confirmation goal, the proposed scheme also reached the goal of only industrial enterprise B, believing that industrial enterprise A saw the final negotiated key.

### B. Formal Analysis of Registration Phase

This section applies the more mature automated protocol analysis software Scyther [41] to demonstrate the security of the proposed scheme. The user writes the authentication negotiation process for sending and receiving messages between roles in Scyther with the security policy definition language. Scyther detects and analyzes security attributes defined by the user. If it searches an attack path, it graphically displays the attack process. Claim events are used to define the security properties to be achieved by the protocol, which mainly

| Scyther results : verify |             |               |               |        |                           |
|--------------------------|-------------|---------------|---------------|--------|---------------------------|
| Claim                    |             |               |               | Status | Comments                  |
| ours                     | KGCX        | ours,KGCX1    | Secret Ah1    | ok     | No attacks within bounds. |
|                          |             | ours,KGCX2    | Secret sigmaA | ok     | No attacks within bounds. |
|                          |             | ours,KGCX3    | Secret PRX    | ok     | No attacks within bounds. |
|                          |             | ours,KGCX4    | Alive         | ok     | No attacks within bounds. |
|                          |             | ours,KGCX5    | Weakagree     | ok     | No attacks within bounds. |
|                          |             | ours,KGCX6    | Niagree       | ok     | No attacks within bounds. |
|                          |             | ours,KGCX7    | Nisynch       | ok     | No attacks within bounds. |
| RoleA                    | ours,RoleA1 | Secret sigmaA |               | ok     | No attacks within bounds. |
|                          | ours,RoleA2 | Secret IDA    |               | ok     | No attacks within bounds. |
|                          | ours,RoleA3 | Secret Ar1    |               | ok     | No attacks within bounds. |
|                          | ours,RoleA4 | Alive         |               | ok     | No attacks within bounds. |
|                          | ours,RoleA5 | Weakagree     |               | ok     | No attacks within bounds. |
|                          | ours,RoleA6 | Niagree       |               | ok     | No attacks within bounds. |
|                          | ours,RoleA7 | Nisynch       |               | ok     | No attacks within bounds. |
| Done.                    |             |               |               |        |                           |

Fig. 4. Verified results of industrial enterprise A with Scyther during the registration phase.

includes Secrecy (Secret) [42], Aliveness (Alive) [43], Weak agreement (Weakagree) [43], Noninjective synchronization (Nisynch) [42], and Noninjective agreement (Niagree) [42]. Secret represents that the parameter items are confidential and secure. Alive represents that the entity is active during the operation of the protocol. Nisynch, Niagree, and Weakagree are mainly used to detect whether the protocol can resist replay attacks, tampering attacks, eavesdropping attacks, and other attacks. References [42] and [43] give a detailed definition.

Scyther generates a separate row for each claim event in the experimental results shown in Figs. 4 and 5 during the registration and authentication key negotiation phases. The first column represents the name of the verification file. The second column represents the role defined by the protocol. The third column shows the claim identifier, and the fourth column represents the claim type and parameters. Through testing the secrecy of  $Ah_1$ ,  $\sigma_A$ , and  $PRX$  generated by KGC and the Alive, Weakagree, Nisynch, and Niagree of this entity, respectively. The verification results in the fifth column indicate that the verification passed. The sixth column further explains that the verification result does not find an attack path within the bounded query, i.e., the proposed scheme is secure. Similarly, the secrecy check results for the partial private key  $\sigma_A$ ,  $IDA$ ,  $Ar_1$  of the role industrial enterprise A also passed and achieved other security properties. Since the interaction process of industrial enterprise A and industrial enterprise B with KGC is the same, this article omits the verification results of industrial enterprise B with key generation center KGC Y. Their verification results are also secure.

### C. Formal Analysis of Authentication Key Negotiation Phase

Fig. 5 shows the results of the proposed scheme with Scyther for verification during the authentication key negotiation phase.  $SA$  and  $SB$  represent the private keys of industrial enterprises A and B, respectively. The  $sk_{ab}$  represents the negotiation key of industrial enterprises A and



| Scyther results : verify |             |                         |        |          |             |
|--------------------------|-------------|-------------------------|--------|----------|-------------|
| Claim                    |             |                         | Status |          | Comments    |
| ours, RoleA              | ours,RoleA1 | Secret SA               | OK     | Verified | No attacks. |
|                          | ours,RoleA2 | SKR skab                | OK     | Verified | No attacks. |
|                          | ours,RoleA3 | Secret Ar3              | OK     | Verified | No attacks. |
|                          | ours,RoleA4 | Secret scalarmul(PB,SA) | OK     | Verified | No attacks. |
|                          | ours,RoleA5 | Secret QAB              | OK     | Verified | No attacks. |
|                          | ours,RoleA6 | Alive                   | OK     | Verified | No attacks. |
|                          | ours,RoleA7 | Weakagree               | OK     | Verified | No attacks. |
|                          | ours,RoleA8 | Niagree                 | OK     | Verified | No attacks. |
|                          | ours,RoleA9 | Nisynch                 | OK     | Verified | No attacks. |
| RoleB                    | ours,RoleB1 | Secret SB               | OK     | Verified | No attacks. |
|                          | ours,RoleB2 | SKR skab                | OK     | Verified | No attacks. |
|                          | ours,RoleB3 | Secret Br3              | OK     | Verified | No attacks. |
|                          | ours,RoleB4 | Secret scalarmul(PA,SB) | OK     | Verified | No attacks. |
|                          | ours,RoleB5 | Secret QBA              | OK     | Verified | No attacks. |
|                          | ours,RoleB6 | Alive                   | OK     | Verified | No attacks. |
|                          | ours,RoleB7 | Weakagree               | OK     | Verified | No attacks. |
|                          | ours,RoleB8 | Niagree                 | OK     | Verified | No attacks. |
|                          | ours,RoleB9 | Nisynch                 | OK     | Verified | No attacks. |
| Done.                    |             |                         |        |          |             |

Fig. 5. Results of verification between industrial enterprise A and industrial enterprise B with Scyther during the authentication key negotiation phase.

B. The session-key reveal (SKR) represents the confidentiality of the session key,  $Ar_3$  and  $Br_3$  represent the random numbers generated by industrial enterprises A and B, respectively. Their secrecy verification results are secure. Second, the  $QAB$  and  $QBA$  sent by both parties are secure. In addition, the verification result of the security of the point multiplication result of the entity's private key and the other party's public key is also secure. Finally, the verification results of the Alive, Weakagree, Nisynch, and Niagree of the two entities are also secure, so the proposed scheme is secure. The verification process found no attacks.

#### D. Security Properties Analysis

In the following, this section analyzes the security properties of the proposed scheme individually.

- 1) *Forward Confidentiality* [44]: This security property means that even after an attacker compromises the entity's private key, it still cannot crack the previously sent messages. Since the negotiation key of the proposed scheme is jointly generated by the entity's private key and a random number,  $sk_{ab} = SA \cdot Ar_3 \cdot rpa = Br_3 \cdot SB \cdot rpb$ , the negotiation key generated each time differs due to the different random numbers  $Ar_3$  and  $Br_3$  included. Even if an attacker knows the entity's private key, it still cannot crack the sent messages.
- 2) *Replay Attack*: The proposed scheme adds a timestamp to each sent message to prevent an attacker from stealing a previous message and resending a key negotiation request to obtain the entity's negotiation key. It performs a mixed hash operation with the timestamp to prevent it from being replaced. For example, industrial enterprise

A contains  $t_1$  in the sent  $Ah_3$  and  $Ah_4$ . Then, industrial enterprise A encrypts  $Ah_4$  with the negotiation key and sends it to industrial enterprise B. After receiving the ciphertext feedback from industrial enterprise B. It verifies whether  $Aciphertext$  and  $Bciphertext$  are equal to determine whether a replay attack has occurred. Industrial enterprise B adopts the same method to verify whether a replay attack has occurred on the sent message.

- 3) *Resisting Eavesdropping Attack*: After receiving a partial private key sent to itself by the KGC, the industrial enterprise will check the correctness of the partial private key and then further process them to generate its public key and private key, (1) shows the proof of the correctness of the partial private key. During the key negotiation process, the proposed scheme keeps the private key and fresh random numbers undisclosed on the network, thereby the proposed scheme enabling it to resist eavesdropping attacks.
- 4) *Man-in-the-Middle Attack*: In the key negotiation process, each entity needs to send a message encrypted with its private key and the receiver's public key. The receiver can only verify the correctness of the message with the sender's public key and the receiver's private key. Thus, the proposed scheme can resist the man-in-the-middle attack. For example, industrial enterprise A utilizes its own private key  $SA$  and random value  $Ar_3$  to generate  $QAB$ . Industrial enterprise B must utilize its own private key  $SB$  and industrial enterprise A's public key  $PA$  to verify that the authentication request is sent by industrial enterprise A after receiving the authentication request, thus avoiding man-in-the-middle attacks.
- 5) *Identity Impersonation Attack* [45]: When industrial enterprise A sends a key negotiation request with industrial enterprise B, since the blockchain stores legitimate entities' identity and public key, the receiver retrieves information such as the sender's public key in the blockchain by invoking a smart contract. Then, the receiver utilizes its private key and the sender's public key to verify whether the sender's request is trustworthy and reliable to resist the identity impersonation attack.
- 6) *Distributed Denial-of-Service Attack*: Since legitimate entities will upload their public keys and identities to the blockchain, its distributed and multicopy nature ensures their availability. The attacker still cannot affect the system operation even if one node is compromised. The entity can obtain the sender's public key through other blockchain nodes for the key negotiation process. In addition, the key negotiation process does not require the participation of the KGC, so the proposed scheme can resist distributed denial-of-service attacks.
- 7) *Decentralization*: The proposed scheme builds on top of the Ethereum consortium blockchain, dividing each entity into different subnets in a network, i.e., network slices. A single entity only needs to interact with the KGC in its region for identity registration. It helps solve the problem of performance bottlenecks due to the trust center handling more operations. Thereby, the proposed scheme achieves decentralization.

- 8) *Resist Key Escrow*: Compared with the previous centralized key generation method, the key generation process in this article generates the applicant's private key jointly by the applicant and the KGC. The KGC cannot infer the applicant's private key. That is, the entity generates its private key by its secret value and partial private key sent to itself by KGC. Only the entity knows its private key, thus avoiding the key escrow problem.
- 9) *Mutual Authentication*: Each entity needs to verify the identity of the sender and the legitimacy of the request with its private key and the other party's public key and utilizes timestamps to prevent replay attacks. After both parties have passed the authentication, they can generate a reliable negotiation key. Since both Aciphertext and Bciphertext contain the public and private keys of the sender and receiver, new random values, timestamps, and other information, the receiver and sender verify whether the encrypted Aciphertext and Bciphertext are equal to achieve mutual authentication.
- 10) *Cross-Domain Authentication*: Each participating entity obtains information such as public keys of entities in other regions by accessing the consortium blockchain. A single KGC is deployed in each domain to take care of the registration process of the entities in the region. The KGCs in different regions verify the entity's identity and the public key, and other information through the PoA consensus mechanism. Entities in different regions authenticate each other for cross-regional authentication, thus improving the capability of cross-regional cooperation.

## VI. PERFORMANCE ANALYSIS

### A. Experimental Environment

This section details the experimental environment of the proposed scheme to facilitate the repetition of the experiments in this article. A virtual machine of Ubuntu 20.04 LTS version on a Dell OptiPlex 7080 PC with Windows 10 via Oracle VM Virtual virtualization software. The PC's processor configuration is Intel Core i5-10500 CPU @ 3.10 GHz and 8G RAM space allocated to the virtual machine. The Go language version of the pairing-based cryptography library [46] is the underlying cryptographic technology support, where the bilinear pairing theory can be found in [47]. The proposed solution uses the Go language to write the main program and deploys the Ethereum consortium blockchain by calling the Go-Ethereum (Geth) package. As Ethereum uses the solidity language to write smart contracts, the abigen tool in the Geth package can compile the smart contracts as Go code for the main program to call. The Goroutine in the Go language can achieve lightweight and highly concurrent programs to meet the high-performance needs of the Industrial Internet. This article constructs the pairing of type A on the elliptic curve  $y^2 = x^3 + x$  in the finite domain  $F_q$ . The bilinear pairing is  $\hat{e} : G_1 \times G_1 \rightarrow G_T$ , where  $\text{rbits} = 160$  and  $\text{qbts} = 512$ . In order to highlight the advantages of the method designed by the cryptographic protocol and ignore the differences of different elliptic curves, the elliptic curve  $y^2 = x^3 + x$  in the

TABLE III  
IMPLICATIONS AND COMPUTATIONAL OVERHEAD OF  
DIFFERENT CRYPTOGRAPHIC OPERATIONS

| Cryptographic operation       | Implication               | Time cost( $\mu\text{s}$ ) | Output length(B) |
|-------------------------------|---------------------------|----------------------------|------------------|
| $T_h$                         | Hash function             | 2                          | 32               |
| $T_{XOR}$                     | XOR operation             | 0                          | 32               |
| $T_{pm}(Z_p^* \times G_1)$    | Point multiplication      | 2096                       | 128              |
| $T_{Add}(G_1 \times G_1)$     | Addition in $G_1$         | 9                          | 128              |
| $T_{exp}(G_1 \times Z_p^*)$   | Exponential in $G_1$      | 2653                       | 128              |
| $T_{Mul}(G_1 \times G_1)$     | Multiplication in $G_1$   | 8                          | 128              |
| $T_{Mul}(G_T \times G_T)$     | Multiplication in $G_T$   | 3                          | 128              |
| $T_{exp}(G_T \times Z_p^*)$   | Exponentiation in $G_T$   | 85                         | 128              |
| $T_{Mul}(Z_p^* \times Z_p^*)$ | Multiplication in $Z_p^*$ | 1                          | 20               |
| $T_{Add}(Z_p^* \times Z_p^*)$ | Addition in $Z_p^*$       | 1                          | 20               |
| $T_{Inv}(Z_p^*)$              | Inversion in $Z_p^*$      | 3                          | 20               |
| $T_p$                         | Pairing operation         | 2046                       | 128              |
| $T_{sen}$                     | AES encryption            | 8                          | 48               |
| $T_{sde}$                     | AES decryption            | 58                         | 32               |
| $T_{kdf}$                     | Key derivation function   | 72064                      | 97               |
| $T_{hmac}$                    | Hmac operation            | 707                        | 88               |

finite field  $F_q$  is adopted uniformly for all the elliptic curves utilized in the comparison schemes.

Comparing the proposed scheme with seven related schemes in the last three years, Table III lists the cryptographic operations involved in these schemes. This article conducts 100 experiments for each listed cryptographic operation to get the average computation consumption time of all cryptographic operations. Table III also presents the number of bytes of the output result of each cryptographic operation. For example,  $T_h$  represents the hash operation. The number of bytes output is 32 bytes, and the computation consumption time is 2  $\mu\text{s}$ .  $T_{pm}(Z_p^* \times G_1)$  represents the multiplication of one element from the group  $G_1$  and the ring  $Z_p^*$ , respectively, and the computation consumes 2096  $\mu\text{s}$ .  $T_{Inv}(Z_p^*)$  represents the inverse of the element in the ring  $Z_p^*$ . In Table III, the point multiplication  $T_{pm}(Z_p^* \times G_1)$ , the exponential operation  $T_{exp}(G_1 \times Z_p^*)$  in the group  $G_1$ , the pairing operation  $T_p$  and the key derivation function (KDF) operation all consume more than 2000  $\mu\text{s}$  of computation time. The hash-based message authentication code  $T_{hmac}$  operation consumes 707  $\mu\text{s}$ . While all other cryptographic operations consume less than 100  $\mu\text{s}$ , the KDF operation consumes the most computation, equal to about 72 ms. The exclusive disjunction (XOR) operation consumes the least time, less than 1  $\mu\text{s}$ . In addition, the key that advanced encryption standard (AES) symmetric encryption algorithm takes is a 128-bit key,  $T_{sen}$  and  $T_{sde}$  stand for AES symmetric encryption algorithm and AES symmetric decryption algorithm, respectively. In terms of output results, the number of bytes output by the cryptographic operations involving the elements in group  $G_1$  are all 128 bytes, and the hash operation, the XOR operation, and the AES symmetric decryption algorithm all output 32 bytes. Moreover, the output results of the operations of the internal elements in  $Z_p^*$  are all 20 bytes. The output results of the KDF operation and the hash-based message authentication code (HMAC) operation are also relatively large, 97 and 88 bytes, respectively.

TABLE IV  
COMPARISON OF COMPUTATIONAL OVERHEAD OF DIFFERENT SCHEMES

| Scheme    | Registration phase                                           | Authentication key negotiation phase                                                       |
|-----------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| PEAS [17] | $2T_{pm}(Z_p^* \times G_1)$                                  | $8T_{pm}(Z_p^* \times G_1) + 2T_{Add}(G_1 \times G_1)$                                     |
| SLAS [16] | $2T_{pm}(Z_p^* \times G_1)$                                  | $10T_{pm}(Z_p^* \times G_1) + 2T_{Add}(G_1 \times G_1) + 2T_{kdf}$                         |
| CTAK [22] | $5T_{pm}(Z_p^* \times G_1)$                                  | $8T_{pm}(Z_p^* \times G_1) + 6T_{Add}(G_1 \times G_1)$                                     |
| BMEC [24] | $4T_{pm}(Z_p^* \times G_1) + T_{Add}(G_1 \times G_1) + 2T_p$ | $4T_{pm}(Z_p^* \times G_1) + 2T_{Add}(G_1 \times G_1) + 2T_p$                              |
| BASA [19] | $4T_{pm}(Z_p^* \times G_1)$                                  | $8T_{pm}(Z_p^* \times G_1) + 2T_{Add}(G_1 \times G_1) + 4T_{exp}(G_T \times Z_p^*) + 4T_p$ |
| EPPA [20] | $9T_{exp}(G_1 \times Z_p^*) + 2T_{Mul}(G_1 \times G_1)$      | $5T_{exp}(G_1 \times Z_p^*) + 4T_{Mul}(G_1 \times G_1) + 6T_p$                             |
| LASG [18] | $3T_{pm}(Z_p^* \times G_1) + T_{sen}$                        | $7T_{pm}(Z_p^* \times G_1) + 3T_{sen} + 3T_{sde} + 4T_{hmac}$                              |
| Ours      | $12T_{pm}(Z_p^* \times G_1) + 4T_{Add}(G_1 \times G_1)$      | $8T_{pm}(Z_p^* \times G_1) + 2T_{sen}$                                                     |

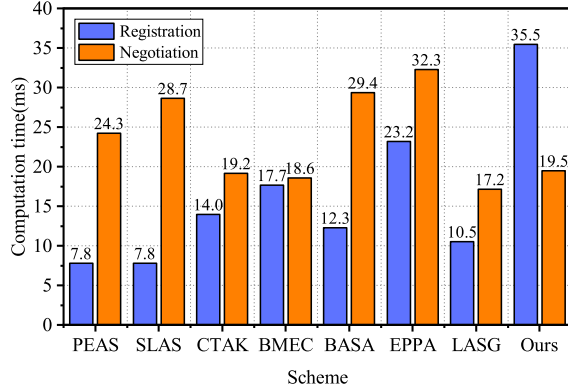


Fig. 6. Histogram of computational costs of relevant schemes.

### B. Computational Cost Analysis

Compared with seven related schemes, Table IV presents statistics of the cryptographic operations consumed by each scheme during the entity registration and the authentication key negotiation phases. The entity registration phase also includes the system initialization process. A virtual machine simulated the other seven schemes. The computational cost histogram and communication cost histogram for all schemes were measured and presented in Figs. 6 and 7, respectively. All data are averages calculated by running 20 experiments. The entity registration phase mainly refers to the interaction between the participating entities and the trust center. The statistics did not include the cryptographic operations in each scheme where the computation consumes less than 8  $\mu$ s. In addition, for the convenience of remembering each scheme, the combination of the first four real-meaning words in the title of each scheme for representation, i.e., [16], [17], [18], [19], [20], [22], and [24] are represented as SLAS, PEAS, LASG, BASA, EPPA, CTAK, and BMEC, respectively. Moreover, the scheme EPPA only considers the registration process of the user and the cloud server at KGC and the authentication process between them.

The time consumption of the proposed scheme in the registration stage is 35.5 ms, which is higher than the other seven schemes. Because each industrial enterprise will perform a point multiplication operation to check whether the received partial private key is correct after obtaining the partial private key from the KGC. After the industrial enterprise sends the public key to the blockchain, the KGC participating in the consensus mechanism will verify whether the user's public key is correct. Only the correct public key will record into

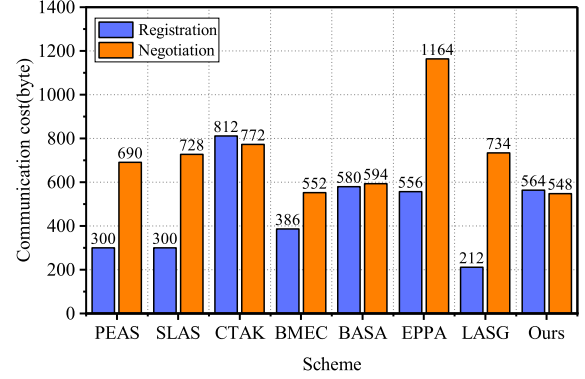


Fig. 7. Histogram of communication cost of related schemes.

the blockchain, so the computational cost consumed is high. The SLAS scheme and PEAS scheme consume the least time because both schemes send their identities to the CA. After that, the CA generates the public key and private key of the corresponding entity and sends the public key and private key to the entity through the secure channel. The EPPA scheme consumes more time for calculation because the registration phase of the EPPA scheme requires nine exponential operations. The BMEC scheme's computational cost is high because the terminal node needs to perform two pairing operations to verify the correctness of the signature after receiving the signature sent to itself by the nearby cache node. The CTAK scheme, the LASG scheme, and the BASA scheme do not have the corresponding verification operation when the entity receives the private key or partial private key from the trust center. Hence, these schemes consume less computation time.

In the authentication key negotiation phase, The proposed scheme mainly performs eight times point multiplication operations and two times AES symmetric encryption algorithms to accomplish cross-domain authentication. The computation time of the proposed scheme is second only to the BMEC scheme and LASG scheme, where the BMEC scheme and LASG scheme are the authentication process between a single entity and the trust center. Since the trust center stores the information about the entity, these two schemes do not require too much computational overhead. Because the EPPA, SLAS, and PEAS schemes need more point multiplication or pairing operations, their computation overhead is more considerable. The SLAS scheme applies KDF operation in key generation, so it consumes the most time. In order to be able to analyze the computation of other parts of this scheme, this article



replaces its KDF operation with a hash operation. The computation of the BASA scheme requires more exponential and pairing operations, so its computational overhead is more considerable. The computational overhead of the CTAK scheme and the proposed scheme are closer. It also needs eight times point multiplication operations. In addition, mapping different amounts of data to the elements in  $Z_p^*$  also consumes a certain amount of computation, and the computation overhead gradually increases as the amount of data increases.

### C. Communication Cost Analysis

Regarding communication cost, the length of the timestamp involved in all relevant schemes is 55 bytes(B), and the name of each entity is two bytes. The length of the element in ring  $Z_p^*$  is 20 bytes. The length of the elements in group  $G_1$  and group  $G_T$  is 128 bytes. In the registration phase, Fig. 7 illustrates the CTAK scheme transmits the most data, including the signature generated by each entity, each entity's public, and KGC's public key. Since it is two entities interacting with KGC in this scheme, the total communication cost is  $2(|ID|(2B) + |T_i|(128B) + |d_i|(20B) + |R_i|(128B) + |P_{pub}|(128B)) = 812B$ , where  $T_i$ ,  $d_i$ ,  $R_i$ , and  $P_{pub}$  are the symbols of the CTAK scheme. The LASG scheme has a minor communication overhead because it only needs to transmit information, such as the public key generated by the NAN for the SM. The LASG and the BMEC schemes consider only the communication process between one entity and the trust center. In contrast, the other schemes must consider the initialization process between two entities and the trust center. The registration process of the PEAS scheme and SLAS scheme is the same, so the communication cost of these two schemes is equal. However, the communication cost of the proposed scheme is less than the CTAK and BASA schemes and redundant to the other schemes. The communication cost of the registration process of the proposed scheme is  $2(|ID|(2B) + |\sigma_A|(20B) + |AR_2|(128B) + |KGC|(2B) + |AR_1|(128B) + |ID|(2B)) = 564B$ .

The EPPA scheme has the highest communication cost in the authentication phase because it has long-signing data. It comprises multiple elements in group  $G_1$  to verify the signature's correctness by pairing operations. The CTAK scheme has the second highest communication cost after the EPPA scheme. The high communication cost is because the elements in group  $G_1$  mainly dominate the interentity interaction process. In addition, the communication cost of the PEAS scheme, SLAS scheme, and LASG scheme is around 700 bytes. However, the BMEC scheme, BASA scheme, and the proposed scheme are below 600 bytes. The proposed scheme consumes a minor communication overhead compared to all other schemes. The proposed scheme includes each entity transmitting 2 bytes to obtain the public keys of other entities through the blockchain. Additionally, each entity transmits 224 bytes to generate the signature of the random number, along with information such as timestamps and hash values. Finally, each entity to send an encrypted message confirmation with the negotiated key, transmitting 48 bytes. Hence, the communication cost of the proposed scheme in the authentication phase is  $2(2 + 224 + 48) = 548$  bytes.



Fig. 8. Blockchain-based Industrial Internet authentication scenario.

### D. Practical Application

The proposed scheme runs on an Ethereum consortium blockchain built by four Raspberry Pi computers to adapt to the Industrial Internet, as shown in Fig. 8. Two Raspberry Pi 4B computers serve as KGCs in each region with Broadcom BCM2711, Quad-core Cortex-A72 (ARM v7l+) 64-bit SoC @1.5 GHz, 8G RAM, and 16G TF card. The other two Raspberry Pi 400 computers simulate the Industrial Internet enterprises in two separate regions. Their configurament is Broadcom BCM2711, Quad-core Cortex-A72 (ARM v7l+) 64-bit SoC @1.8 GHz, 4G RAM, and 16G TF card. All devices are communicating in two subnets generated by the mobile hotspot. A Raspberry Pi 400 computer and a Raspberry 4B computer belong to one subnet. The blockchain data storage and retrieval smart contract are written and compiled in Remix with Solidity language. Then, the smart contract is deployed on a consortium blockchain built by Geth installed by each node, which applies the PoA consensus mechanism. The PoA consensus mechanism is a reputation-based consensus algorithm. Nodes become verifiers by pledging the value of their identity, and the process of becoming a verifier is subject to rigorous auditing. The PoA consensus mechanism has better scalability and lower transaction latency than the PoW consensus mechanism [48]. All KGCs participate in the consensus process. The consortium blockchain generates a block every 10 s. All devices interact with the consortium blockchain's Geth by running the Go language program of the proposed scheme.

After 20 experiments, the time of the registration phase to be 11.16 s, and the computational overhead of the authentication key negotiation phase to be 0.16 s. In the authentication phase, the communication cost for the KGC uploading its identity and public key to the blockchain is  $|KGC|(2) + |PKX|(128) = 130$  bytes. Moreover, the communication cost for industrial enterprise A to obtain the public key of the KGC from the blockchain is  $|PKX|(128) = 128$  bytes. After the consensus mechanism, the communication cost of uploading industrial enterprise A's public key and identity is  $|PA|(128) + |IDA|(2) = 130$  bytes. In the authentication key negotiation phase, the communication cost of industrial enterprise A to obtain the public key of industrial enterprise B is



TABLE V  
COMPARISON OF SECURITY PROPERTIES OF DIFFERENT SCHEMES

| Security properties                  | BMEC [24] | SLAS [16] | CTAK [22] | PEAS [17] | BASA [19] | EPPA [20] | LASG [18] | Ours |
|--------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|------|
| Anonymity                            | ✓         | ✓         | ×         | ✓         | ✓         | ×         | ✓         | ×    |
| Mutual authentication                | ×         | ✓         | ×         | ✓         | ✓         | ✓         | ✓         | ✓    |
| Session key establishment            | ×         | ✓         | ×         | ✓         | ✓         | ×         | ✓         | ✓    |
| Decentralization                     | ✓         | ×         | ×         | ×         | ✓         | ×         | ×         | ✓    |
| Resist key escrow                    | ✓         | ×         | ✓         | ×         | ×         | ✓         | ×         | ✓    |
| Revocable                            | ×         | ✓         | ✓         | ✓         | ×         | ✓         | ✓         | ×    |
| Cross-domain authentication          | ✓         | ×         | ×         | ×         | ✓         | ✓         | ×         | ✓    |
| Forward confidentiality              | ✓         | ✓         | ✓         | ✓         | ✓         | ✓         | ✓         | ✓    |
| Resist eavesdropping attack          | ×         | ×         | ×         | ×         | ×         | ✓         | ✓         | ✓    |
| Resist replay attack                 | ×         | ✓         | ×         | ✓         | ×         | ✓         | ✓         | ✓    |
| Resist man-in-the-middle attack      | ×         | ✓         | ×         | ×         | ×         | ✓         | ✓         | ✓    |
| Resist identity impersonation attack | ✓         | ×         | ×         | ×         | ✓         | ✓         | ✓         | ✓    |
| Resist DDoS attack                   | ✓         | ×         | ×         | ×         | ✓         | ×         | ×         | ✓    |

$|PB|(128) + |IDB|(2) = 130$  bytes. So the communication cost associated with the blockchain in the registration and authentication key negotiation phases is  $2(130 + 128 + 130) = 776$  bytes and  $2 \times 130 = 260$  bytes, respectively.

#### E. Comparative Analysis of Security Properties

In this section, the types of attacks and security properties that the proposed scheme resists with other related schemes are presented in Table V, where ✓ represents support for security features and × represents no support for security features. Regarding functional security features, the CTAK scheme, the EPPA scheme, and the proposed scheme do not implement anonymity. The proposed scheme needs to disclose its identity information in the consortium blockchain to facilitate industrial companies in other regions to contact itself. The BMEC scheme achieves only single-direction authentication rather than bidirectional authentication. The CTAK scheme does not validate the sender's information, so the scheme does not implement single-way authentication. The BMEC, CTAK, and EPPA schemes do not implement session key establishment or confirmation functions. The LASG, BASA, PEAS, and SLAS schemes generate the entity's public and private keys from the trust center, so these schemes do not resist to the key escrow problem. The BMEC scheme, BASA scheme, and the proposed scheme adopting a trusted authentication platform based on blockchain, the entity's public key cannot be modified after being published to the blockchain, so these mechanisms cannot achieve the public key revocability function. Participants can revoke the transaction by publishing the public key on the blockchain and regenerating a new one for the network-wide query. Furthermore, the SLAS, CTAK, LASG, and PEAS schemes do not achieve cross-domain authentication.

In terms of resisting malicious attacks, all schemes achieve forward secrecy. The EPPA scheme and the proposed scheme use a certificate-free cryptographic mechanism. The entity can check the correctness of the partial private key after receiving the partial private key sent to itself by the trust center and then operate further to generate its public and private key. So these two schemes can resist eavesdropping attacks. In addition, NAN generates the key of the SM device in the LASG scheme on the device and then

deploys it to the application site so that this scheme can resist the eavesdropping attack. Other schemes send the key to the entity through the secure channel, and the entity does not process the private key further, so these schemes cannot resist the eavesdropping attack. The BMEC, CTAK, and BASA schemes lack corresponding timestamp checking or new value confirmation mechanisms, so these schemes cannot resist replay attacks. The BMEC, CTAK, PEAS, and BASA schemes lack authentication or challenge-response mechanisms for the sender's identity, so these schemes cannot resist man-in-the-middle attacks. The SLAS, CTAK, and PEAS schemes lack an association or verification relationship between identity and public key information, so these schemes cannot resist identity impersonation attack. The SLAS, CTAK, PEAS, EPPA, and LASG schemes are responsible for the registration process of entities by a single trust center. So these trust centers are vulnerable to distributed denial-of-service attacks and do not achieve decentralized authentication purposes.

Through the above comparative analysis of security attributes, this article can conclude that the EPPA mechanism is unsuitable for the confidential transmission of large amounts of data between industrial enterprises across regions because it only achieves mutual authentication and does not achieve session key establishment. Because public key encryption is only suitable for the confidential transmission of a small amount of data and the symmetric encryption algorithm is suitable for the confidential transmission of a large amount of data, the key negotiation process is to generate the key for the symmetric encryption algorithm. In addition, the single KGC in the EPPA scheme is responsible for the public-key registration of the whole network entities, so the KGC is vulnerable to distributed denial-of-service attacks, which causes the whole network to collapse. The BMEC scheme only realizes one-way authentication and does not realize session key establishment, so the mechanism cannot be suitable for the confidential transmission of a large amount of data between industrial enterprises across regions.

The PEAS and SLAS schemes comprise a single CA responsible for generating public and private keys for SMs and NAN gateways across the network. The CA has certificate management complexity and key escrow problems. Hence, these two mechanisms are not suitable for the Industrial Internet. The neighborhood area gateways in the LASG

scheme generate authentication information in their devices before SM deployment, so this scheme is unsuitable for cross-domain authentication and data transfer. The BASA scheme is less secure in the Industrial Internet because of its key escrow problem and inability to resist replay attacks. The CTAK scheme is unsuitable for the Industrial Internet environment because it is not resistant to replay and man-in-the-middle attacks. The proposed scheme implements functions such as cross-domain authentication and key negotiation. However, the proposed scheme is also resistant to replay attacks, man-in-the-middle attacks, and identity imitation attacks. Hence, the proposed scheme is suitable for the Industrial Internet environment.

## VII. CONCLUSION

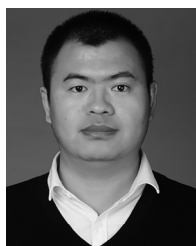
This article presents a blockchain-based certificate-free cross-domain authentication mechanism for Industrial Internet. The consortium blockchain is a trusted platform for industrial enterprises to publish information and system parameters to construct a trusted society. The proposed scheme achieves the intended authentication goal in SVO logic proof, and Scyther verifies the security of the proposed scheme. Compared with other related schemes regarding computational cost and communication overhead. The experimental results show that the proposed scheme has higher communication and computational costs in the registration phase but lower computational cost and communication overhead in the authentication key negotiation phase. Since the registration process is rare in industrial enterprises, the proposed scheme meets the realistic application requirements. The proposed scheme deploys in an industrial consortium blockchain built by Raspberry Pi computers to demonstrate its feasibility. Finally, the results proved the reliability of the proposed scheme through a comparative analysis with other related schemes regarding security properties.

In future work, to further improve the scalability and transaction throughput of the Ethereum consortium blockchain, we will study the blockchain architecture based on sharding to improve the scalability and throughput of the blockchain. We will integrate more cryptographic algorithms to achieve highly reliable data security protection to meet Industrial Internet enterprises' diverse privacy protection requirements. In addition, to extend the functionality of the proposed scheme, we will study the certificate-free cross-domain access control mechanism in the Industrial Internet environment. By adopting flexible access control methods, users achieve security control and intelligent data transmission services for different industrial devices.

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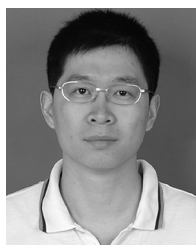
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